Pulmonary Function Tests (PFT)

Last updated March 4, 2026 20 min read ScholarRx

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Adapted from: Pulmonary Function Tests

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Explain the purpose of pulmonary function testing.
2. Define how to measure lung volumes and capacities.
3. Differentiate between subtypes of obstructive and restrictive lung diseases using PFT findings.
4. Explain the use of the diffusion capacity of the lungs for carbon monoxide (DLCO).

CASE CONNECTION

A 68-year-old man with a 45-pack-year smoking history presents with gradually worsening shortness of breath over the past several years. He reports a chronic dry cough and difficulty keeping up with his daily walks. Physical exam shows a thin-appearing man with **barrel-shaped chest** and **decreased breath sounds** bilaterally. Chest X-ray reveals **hyperinflated lungs with flattened diaphragms**. The clinician orders pulmonary function testing.

GO TO CONCLUSION

What Are Pulmonary Function Tests? (LO1)

How do we objectively assess how healthy a patient's lungs are? Pulmonary function tests (PFTs), in addition to the history, physical exam, and radiology studies, are a noninvasive way to help determine overall lung health. These tests determine lung volumes and capacities, rate of air flow, and the quality of alveolar gas diffusion. Because they can help us make a diagnosis and determine effective management, these common tests are a routine part of pulmonary medicine.

There are a variety of situations where PFTs are indicated. These include patients with:

- Persistent cough
- Dyspnea
- Chronic respiratory disease, to follow the progress of disease and effectiveness of treatment

In this brick, we will start by outlining the various lung volumes and capacities that can be measured; explain methods of measurement, including spirometry; describe abnormal spirometry patterns; then explain how to measure diffusing capacity.

Pulmonary function tests are indicated for the assessment of cough and dyspnea and to evaluate patients suspected of having chronic lung disease.

What Are Lung Volumes and Capacities? (LO2)

PFTs measure lung volumes and capacities. What are these? They are the different measurements of the amount of air in the lungs during the phases of the respiratory cycle. The most common ones are shown in Figure 1.

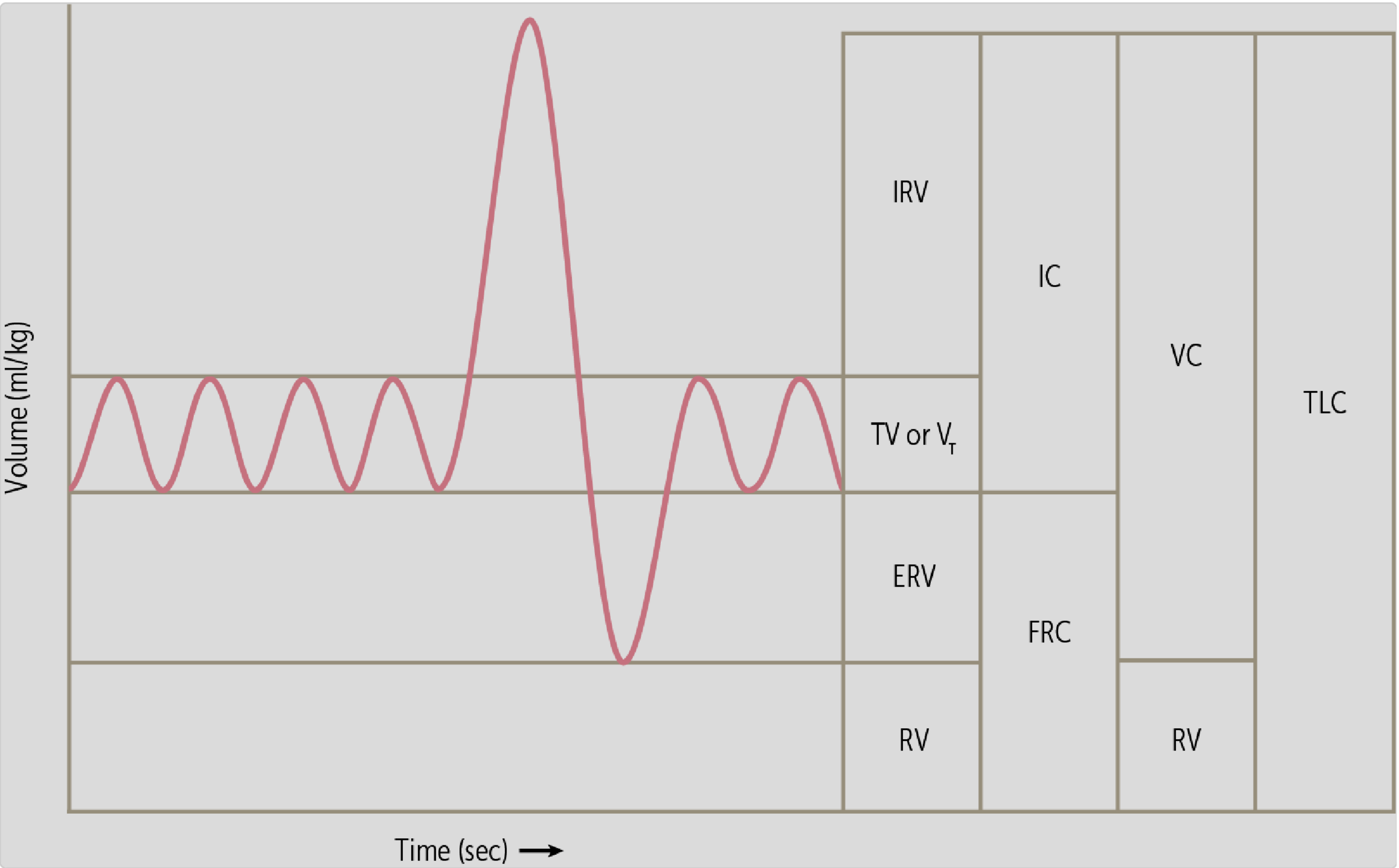

Figure 1 Lung volumes and capacities

Lung Volumes

Knowing the lung volumes can help us evaluate obstructive and restrictive lung disorders. The four basic lung volumes are:

- Tidal volume (**TV**) is the normal amount of air—about half a liter—that you breathe in or out during a single respiration during quiet breathing. This is the ordinary breathing of life.
- Inspiratory reserve volume (**IRV**) is the additional air that you can inhale or inspire above tidal volume during a deep inspiration—for example, the extra air you'd inhale before diving into water.
- Expiratory reserve volume (**ERV**) is the additional air that you can exhale during a forced expiration—think of the extra exhale of blowing

out all the candles on a birthday cake with only one breath.

- **Residual volume** (**RV**) is the air that remains in the lungs no matter how forcefully you blow out. This residual volume exists because the chest wall cannot collapse any further, leaving some air in the alveoli. The residual volume is actually useful, because it helps prevent the collapse of the alveoli (atelectasis).

Lung Capacities

The term “lung capacities” refers to combinations of two or more lung volumes. The four standard lung capacities are:

- **Vital capacity** (**VC**) is all the air that can be expired after maximal inspiration; it is the sum of tidal volume, inspiratory reserve volume, and expiratory reserve volume. This is routinely measured during spirometry (see below).
- **Inspiratory capacity** (**IC**) is all the air that can be inhaled; it is the sum of the tidal volume and the inspiratory reserve volume.
- **Functional residual capacity** (**FRC**) is the air that remains following a normal expiration; it is the sum of the expiratory reserve volume and the residual volume.
- **Total lung capacity** (**TLC**) is the maximum amount of air that the lungs can hold; it is the sum of all four lung volumes (or the vital capacity plus the residual volume). In a healthy adult the TLC is approximately 6 liters. It’s approximately 2 to 4 liters in children.

The vital capacity is the amount of air that can be expired after maximal inspiration; it can be calculated by adding tidal volume, inspiratory reserve volume, and expiratory reserve volume.

How Do We Measure Lung Volumes and Capacities? (LO2)

There are several ways to measure the above lung volumes and capacities. The most often used and most useful test for routine clinical situations is **spirometry**.

Spirometry

Spirometry measures the **volume of air expired during a forceful and complete expiration**, after a maximal inspiration. Spirometry yields a measurement of vital capacity, but not the other volumes and capacities listed above. It also produces a measure of airflow, the **forced expiratory volume in 1 second** (**FEV_1**). These measurements are then compared to the normal values expected to be seen in a healthy person of the same age, sex, and height.

Note that residual volume cannot be measured using spirometry because it is the air that remains in the lung after a maximal expiration. Measuring residual volume requires more specialized testing, commonly called "full

PFTs". Since the total lung capacity = the vital capacity + residual volume, we cannot measure the total lung capacity with spirometry.

Conducting Spirometry

Spirometry requires expert coaching for the patient to perform it accurately. Let's walk through it step by step.

First, a nose clip is used to seal the nostrils to prevent air leakage from the nose. Then the patient must place their lips tightly around the spirometer mouthpiece and breathe normally for a few breaths. After that, the patient is asked to inhale as deeply as possible and then exhale as hard as possible (Figure 2). This process is repeated at least three times until acceptable and reproducible results are recorded.

Figure 2 Spirometry

Two graphs are produced in a spirometry test: a volume-time curve and a flow-volume loop.

Volume–Time Curve

The volume-time curve plots the volume of exhaled air over time during a forceful expiration (Figure 3).

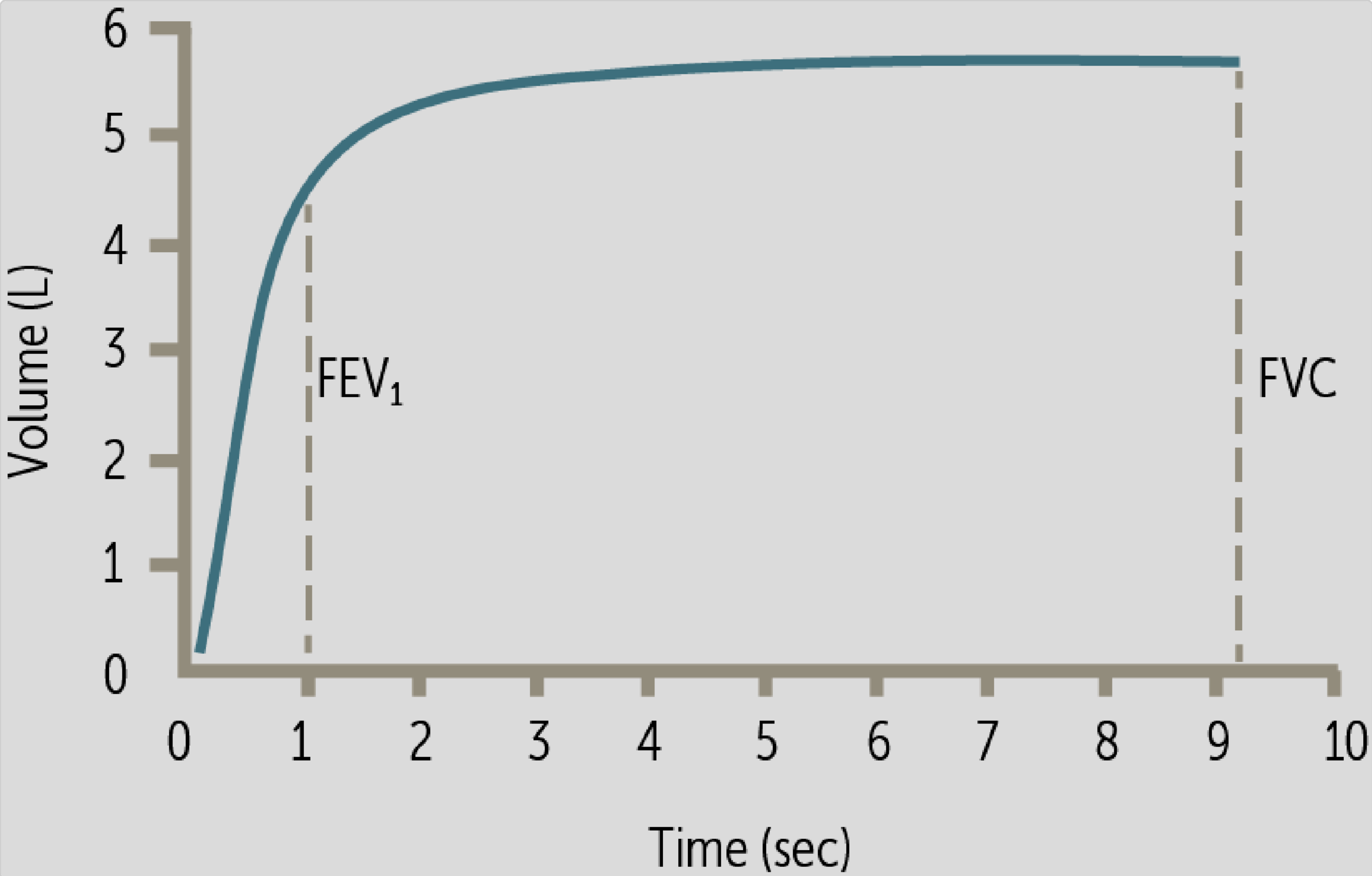

Figure 3 Volume-time curve

Using this curve, we can calculate the three most important numeric values of spirometry:

- **Forced vital capacity (FVC)**, which is the same as vital capacity (all the air that can be expired after maximal inspiration). It's called "forced" here by convention, since it occurs during a forced expiration.
- **Forced expiratory volume** in the first second of expiration (FEV_1).
- **FEV_1/FVC ratio**, which represents what fraction of the forced vital capacity can be exhaled during the first second of a forced expiration. This ratio is a measure of how fast a patient can empty their lungs. This

ratio is very useful in differentiating respiratory disorders (see discussion below).

CLINICAL CORRELATION

What are the normal values? A healthy person has FEV_1 and FVC values that are above 80% of their predicted value for age, sex, and height, and a FEV_1/FVC ratio that is at or above 0.7.

To adjust for age, sex, and height *and avoid using the above, arbitrary fixed ratios*, it is more accurate to compare the results to a healthy age-matched population and identify if the patient is a statistical outlier. This approach uses " **< LLN**" (less than the Lower Limit of Normal, defined as the 5th percentile). This value is also expressed as: **Z score < -1.65**, meaning the value is 1.65 standard deviations below the mean, placing it again below the 5th percentile. For all the fixed ratios described below, one of these two notations is more commonly used currently.

Flow–Volume Loops

A flow-volume loop shows airflow (measured as L/sec) relative to lung volume (in liters) during respiration. Unlike the volume-time curve, there is no plot of time. This value is routinely obtained during spirometry.

Looking at a normal **flow-volume loop**, we see volume displayed on the x-axis and flow displayed on the y-axis (Figure 4).

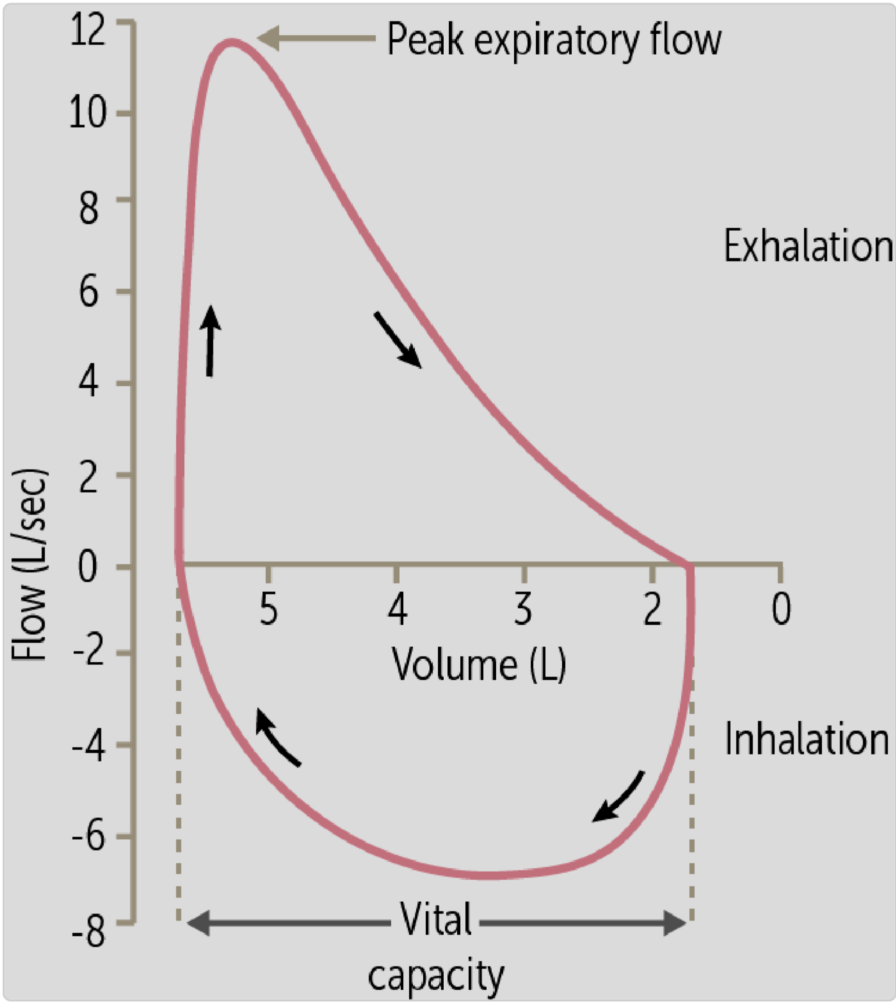

Figure 4 Normal flow-volume loop

The left side of the curve represents maximal inhalation (note the positive lung volume of approximately 5.5 liters there), while the right side depicts maximal exhalation (note that the lung volume is zero). Also observe that the x-axis is "backward," appearing (lower values to the right) because lung volume decreases as the patient exhales.

As the patient inspires (lower curve, right to left) air is drawn into the lungs, causing negative flow on the curve (as air is inspired). At the end of inspiration at far left, the lung volume is maximal and the flow rate becomes 0. Expiration then begins, and the flow rate becomes positive.

Look at the expiratory curve. During exhalation the flow rate quickly increases to the **peak expiratory flow rate**, the greatest flow of air during exhalation. The flow then begins to decrease as air is forced through small airways, which can collapse at low volumes. As we will see, the shape of this curve is very useful in lung diagnosis.

Measuring Other Lung Volumes and Capacities

Spirometry is the most common pulmonary function test and provides the FVC and FEV_1. To obtain residual volumes and TLC, Full PFTs are needed which use additional techniques, either:

- Gas dilution: the patient breathes in an inert gas (eg, helium or nitrogen) and the gas content is measured in the inspired and expired air until there is equilibrium between the two.
- Body plethysmography involves measuring gas volumes as the patient breathes within a closed box (rather like a telephone booth).

The most important numeric values of spirometry are forced expiratory volume in the first second (FEV_1), forced vital capacity (FVC), and the FEV_1/FVC ratio.

How Is Spirometry Used to Diagnose Respiratory Disease? (LO3)

Types of Abnormal Results

Respiratory disorders are often characterized as obstructive, restrictive, both, or neither. We'll now look at how spirometry results and changes in flow-volume loops can help us make these distinctions. Typical flow-volume loops are shown in Figure 5A. The diagnostic algorithm can be referenced through the stepwise measurements Figure 5B.

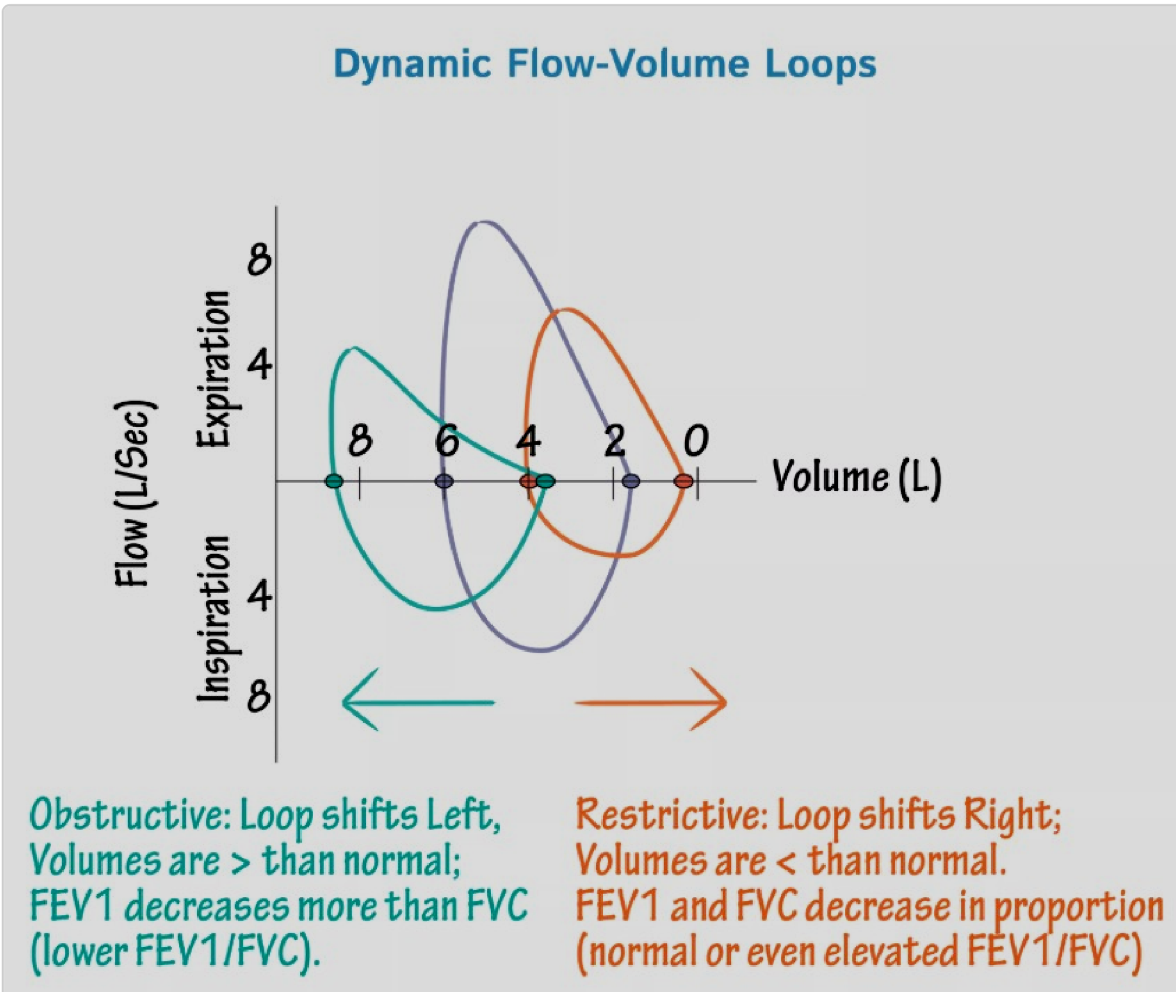

Figure 5A. Flow-Volume Loops

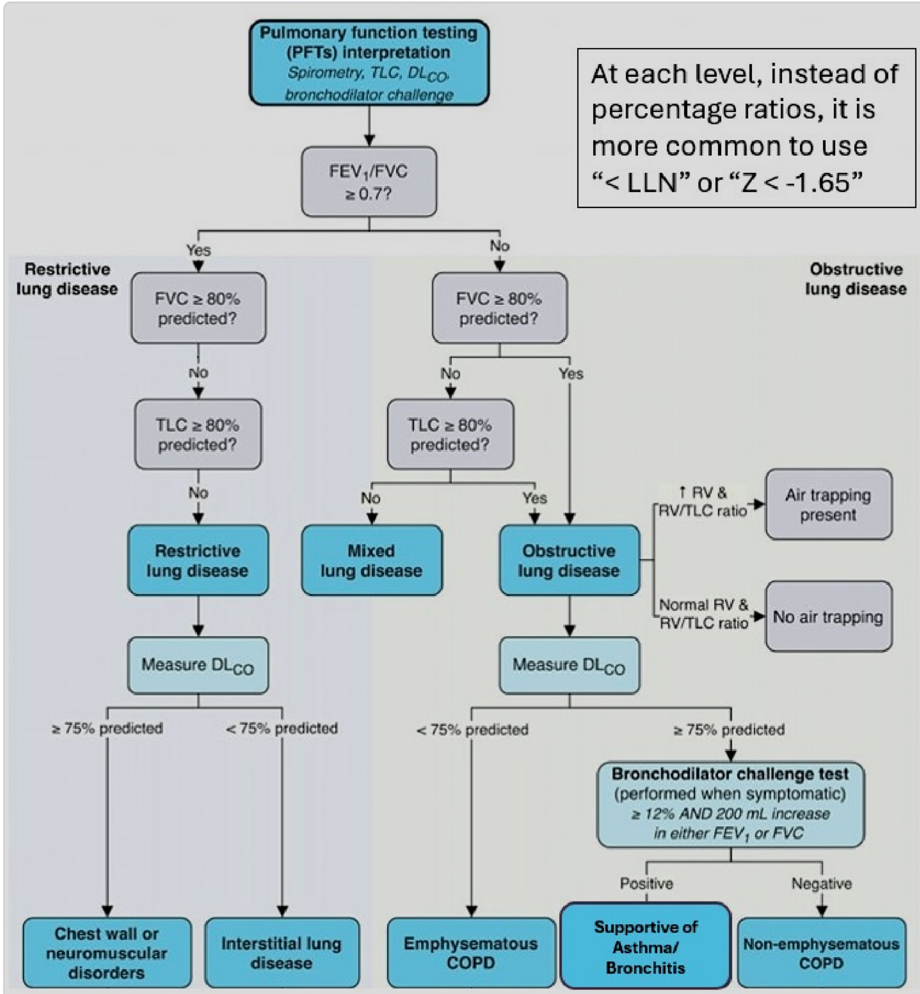

Figure 5B. PFT Algorithm

Obstructive Lung Diseases. These diseases develop when the air cannot easily leave the lungs. Examples include asthma and chronic obstructive pulmonary diseases (COPD) like emphysema and chronic bronchitis. Obstructive diseases create a shortened, concave (scooped out) expiratory curve (see Figure 5, left).

During inspiration, expanding alveoli pull the small airways open; during expiration, collapsing alveoli have less elastic recoil on the airways, worsening any pathologic obstruction. Obstructive lung diseases therefore often increase RV and TLC because air is trapped in the lungs and cannot be expired. **Air trapping** explains the leftward shift in the curve (higher volumes). Over time total lung volume will increase, giving rise to the hyperinflated lungs seen on the physical exam and chest x-ray of patients with COPD.

Also note the drop in peak expiratory flow rate (peak of upper curve), expected in a disease that limits expiratory flow. In spirometry testing, this means that obstructive disease **lowers FEV_1**. The lower FEV_1 along with a mildly decreased or normal FVC results in a **decreased FEV_1/FVC ratio of less than 0.7.**

Restrictive Lung Diseases. These diseases occur when lung pathology makes it difficult for air to enter the lungs, restricting the volume of the lungs. Examples include sarcoidosis, pulmonary fibrosis, and silicosis. FVC and TLC are reduced, creating a narrow, convex expiratory curve that shifts to the right (Figure 5).

There is also usually a drop in expiratory flow, but not as much as in obstructive disease. Expiratory flow depends on lung volume. Smaller lungs generate less elastic recoil and lower driving pressure, leading to reduced expiratory flow despite normal airway resistance (no obstruction). The FEV_1/FVC ratio is greater than or equal to 0.7 (unlike the decreased value in obstructive disease), because both FEV_1 and FVC will decrease to roughly the same degree. The FEV_1 is appropriate for the smaller volume.

Restrictive lung diseases are suggested by an FVC <80% of predicted. Confirmation of restrictive disease, however requires a TLC: <80% of predicted. This requires Full PFTs to obtain the Residual Volume, which is measured by gas dilution (He/N_2) or plethysmography.

Table 1 summarizes the key spirometry differences between obstructive and restrictive lung diseases.

Table 1 Obstructive vs restrictive lung diseases

	Obstructive lung disease	Restrictive lung disease
RV	↑	↓
FRC	↑	↓
TLC	↑	↓
FEV_1	↓↓	↓
FVC	↓	↓
FEV_1/FVC	↓ FEV_1 decreased more than FVC	Normal or ↑ Both FEV_1 and FVC decrease proportionately

Obstructive lung diseases cause a ratio below 0.7, and restrictive lung diseases lead to a normal or mildly increased ratio greater than or equal to 0.7.

Reversibility

Reversibility testing is recommended when the clinician wants to support the diagnosis of asthma. After confirming the obstructive disease, the response to bronchodilation is assessed. If the **FEV_1 improves by >= 12% and 200 ml** from baseline values after bronchodilator therapy, reversibility is confirmed. **This is highly suggestive of Asthma**, but not diagnostic because many patients with COPD also have reversibility**.**

Bronchoprovocation Testing

When obstruction is suspected but baseline spirometry is normal, bronchoprovocation testing (methacholine challenge) is used in an attempt to uncover intermittent obstruction due to airway hyperreactivity. Airway hyperreactivity is also supportive, but not diagnostic of asthma. Reversibility can then be checked to confirm asthma, and improve symptoms after the provocation.

Baseline spirometry is performed, then a very small dose of medication that causes bronchoconstriction—such as **methacholine** (binds to M3 muscarinic receptors) or histamine (binds to H1 histamine receptors)—is administered, and then spirometry is repeated. For a positive result, the medication causes bronchoconstriction provoking asthma symptoms (eg, wheezing) and **at least a 20% drop in FEV_1**. To reverse the effects of the provoking drug, an antagonizing medication such as the beta-2 receptor agonist albuterol is administered.

CLINICAL REASONING

PFTs support the diagnosis of asthma but cannot confirm it alone because of the variable nature of the disorder and overlap with other disorders (such as COPD) that can also have reversibility.

Asthma = chronic airway inflammation + variable symptoms + variable expiratory airflow limitation.

The diagnosis is **based on history, triggers, and symptoms**, not solely on PFTs.

How Do We Measure Diffusing Capacity of the Lungs? (LO4)

The tests described so far all measure lung volumes, capacities, and airflow. Now we'll discuss the **diffusion capacity of the lungs for carbon monoxide** (**DLCO**), a pulmonary function test that can estimate how well gas exchanges across the alveolar membrane, a physiologic parameter often decreased in respiratory disease.

To perform the test, the patient inhales a small amount of administered carbon monoxide (CO) (not enough to cause symptoms!). CO levels are measured in both the inspired and expired air, then the exhaled CO value is subtracted from the inhaled value. Because hemoglobin has a strong affinity for CO, it can be assumed that the difference between inspired and expired CO is the amount of CO that diffuses into the blood from the alveoli. Smaller values mean reduced diffusing capacity, as occurs in several diseases:

- In **obstructive disease like COPD, a DLCO of < LLN confirms emphysema** (which involves alveolar wall destruction and loss of diffusion capacity). A preserved DLCO is characteristic of the chronic bronchitis phenotype of COPD.

- In restrictive disease, a **low DLCO combined with low lung volumes and preserved FEV1/FVC suggests intrinsic pulmonary disease** such as interstitial lung disease such as Idiopathic Pulmonary Fibrosis (IPF), because the alveolar-capillary membrane is thickened or destroyed. **Low lung volumes with normal DLCO instead suggests restrictive disease due to extrapulmonary disorders** such as neuromuscular disorders and chest wall abnormalities (e.g. Myasthenia Gravis, kyphoscoliosis).

CLINICAL CORRELATION

IPF is a restrictive lung disease characterized by the progressive fibrosis of the peribronchiolar interstitium, limiting lung capacity. Spirometry will show a global reduction in lung volumes. These fibrotic changes also increase the distance between the air held in the alveoli and the capillaries of the lungs. This in turn reduces the ability for gas to diffuse out of the alveoli, lowering the DLCO.

Upper Airway Obstructions (LO3)

PFTs may occasionally identify partial upper airway obstructions in patients with dyspnea. They are primarily detected by the altered shape of the flow volume loops (Figure 6). Variable obstructions affect only inspiration or expiration. **Variable extrathoracic obstruction,** such as vocal cord dysfunction, will have a flattened inspiratory limb because the decreased pressure in the airway will pull the airway walls inward. The patient may have stridor during inspiration. **Variable intrathoracic obstruction**, such as tracheomalacia, will have a flattened expiratory limb because increased thoracic pressure during exhalation can narrow the airway. **Fixed obstruction**, such as tracheal stenosis or a large goiter compressing airway will have flattening of both the inspiratory and expiratory limbs.

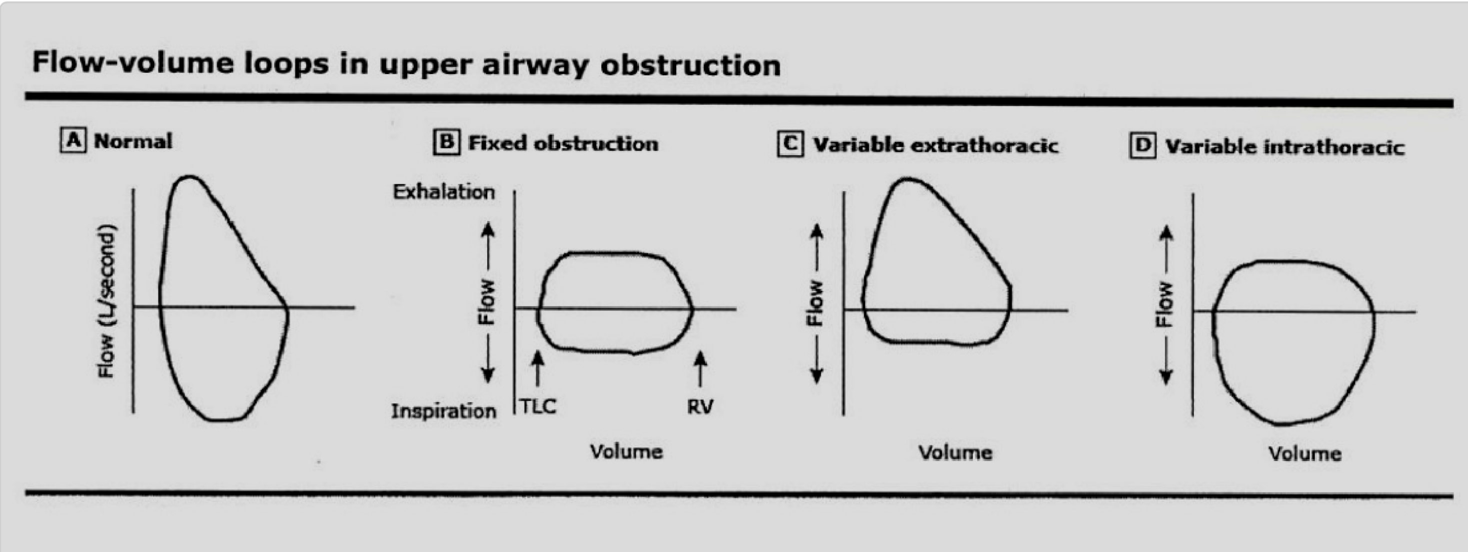

Figure 6. Upper airway obstructive patterns

Effects of Aging

With advancing age, pulmonary function changes reflect structural remodeling of the lung and chest wall. FEV_1 declines disproportionately to FVC as a consequence of reduced elastic recoil and loss of alveolar attachments (radial traction of airways), resulting in premature closure of small airways during forced expiration and a modest age-related reduction in the FEV_1/FVC ratio. Residual volume and functional residual capacity increase owing to air trapping, while vital capacity decreases. Total lung capacity is generally preserved because increased lung compliance from elastin loss is counterbalanced by reduced chest wall compliance related to skeletal stiffening. Diffusing capacity and arterial oxygen tension decline with age due to decreased alveolar surface area, reduced pulmonary capillary blood volume, and greater ventilation–perfusion inequality, whereas arterial carbon dioxide tension remains normal, reflecting intact ventilatory regulation.

DLCO testing identifies intrinsic disorders of the lungs that destroy or thicken the alveolar-capillary membrane.

CASE CONNECTION

BACK TO INTRODUCTION

Spirometry reveals: FEV_1/FVC: 0.60 (< LLN) FEV_1: 55% predicted (< LLN). To further evaluate, Full PFTs are obtained revealing: TLC: 120% predicted, RV: 150% predicted, DLCO: 45% predicted (< LLN).

You interpret these results as an obstructive disorder due to the low FEV_1/FVC. There is no restrictive component: in fact his lungs are hyperinflated due to air trapping which is common in obstructive disorders due to the loss of alveolar recoil on the airways during exhalation. The low diffusion capacity (DLCO) reveals this to be the emphysema subtype of COPD.

Summary

- Pulmonary function tests (PFTs) are noninvasive tests that evaluate lung function.
- PFTs are indicated for the assessment of patients having respiratory symptoms and to follow the progress of patients with respiratory disorders.
- Lung volumes include residual, expiratory reserve, tidal, and inspiratory reserve volumes.
- Lung capacities include functional residual, inspiratory, vital, and total lung capacities.
- The total lung capacity is the sum of the vital capacity and the residual volume.
- Spirometry is the primary test used to assess lung function and measures the volume of air expired at specific points of time during a forceful and complete expiration after a maximal inspiration.
- The most important numeric values of spirometry are the forced expiratory volume in the first second (FEV_1), forced vital capacity (FVC), and the FEV_1/FVC ratio.
- The two main graphs derived from spirometry measurements are the volume-time curve, which plots the volume of exhaled air over time, and the flow-volume loop, which shows airflow relative to lung volume.
- Lung volumes beyond the vital capacity (VC) can be measured using complete PFTs.
- Obstructive lung diseases cause a reduced FEV_1/FVC, and left-shifted, short, concave expiratory portion of the flow-volume loop.
- When asthma is suspected based on initial abnormal spirometry showing obstructive results (FEV_1/FVC LLN), a bronchodilator reversibility test is conducted to support the diagnosis.

- If spirometry is normal but asthma is still suspected, bronchoprovocation testing using methacholine challenge is used to test for bronchial hyperreactivity, common in asthma; a positive test shows a 20% decrease in FEV_1.
- Restrictive lung diseases cause a reduced FVC, normal-to increased FEV_1/FVC, and a right-shifted, narrow, convex expiratory portion of the flow-volume loop; confirmation of restrictive disease requires a TLC < LLN.
- Diffusion capacity of the lungs for carbon monoxide (DLCO) is a pulmonary function test that measures the efficiency of gas exchange.
- DLCO is decreased in several conditions, including emphysema, interstitial lung disease, and many pulmonary vascular diseases.

Review Questions

1. **A 65-year-old patient comes to the clinic for assessment of previously diagnosed lung disease, and spirometry testing is performed. Which of the following lung volumes is a key measure obtained during a forced expiratory maneuver during spirometry?**

   A. Expiratory reserve volume

   B. Functional residual capacity

   C. Inspiratory reserve capacity

   D. Residual volume

   E. Vital capacity

Explanation (requires correct answer)

2. **After spirometry is performed on a patient, the flow-volume loop shows a decreased flow in the effort-dependent segment of expiration. Which of the following values will likely be decreased in this patient?**

A. FEV_1

B. FRC

C. RV

D. TLC

E. TV

Explanation (requires correct answer)

3. **A 35-year-old patient comes to the clinic to be evaluated for dyspnea. Which of the following PFT measurements is consistent with Idiopathich Pulmonary Fibrosis?**

   A. DLCO > LLN

   B. FEV1/FVC < LLN

   C. FVC > LLN

   D. TLC < LLN

Explanation **(requires correct answer)**

and he is discharged home. Spirometry is performed the following week and reveals normal results. If bronchoprovocation is performed, which of the following would be most supportive of the diagnosis of asthma?

A. 20% decrease in FEV_1 after methacholine

B. 20% decrease in FVC after methacholine

C. A drop in blood oxygen saturation after methacholine

D. A rise in blood PCO_2 after methacholine

E. FEV_1/FVC ratio Z score > -1.65 prior to methacoline

Explanation (requires correct answer)

5. Which pulmonary function test is used to measure the efficiency of alveolar gas exchange?

A. Bronchoprovocation test

B. DLCO

C. Gas dilution technique

D. Spirometry

Explanation (requires correct answer)

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 Explain the purpose of pulmonary function testing.

02 Define how to measure lung volumes and capacities.

03 Differentiate between subtypes of obstructive and restrictive lung diseases using PFT findings.

04 Explain the use of the diffusion capacity of the lungs for carbon monoxide (DLCO).

Objective Confidence: 0/4